

Lab Assignment 3

[15/09/2026]

[Conditional Statements and Control Statements]

- Q.1) Write a python program to sum two user input integers. However, if the sum is between x to y it will return True. X and Y is from user.
- Q.2) Write a program that accept a lowercase/ uppercase character from the user and check whether the character is a vowel or consonant.
- Q.3) Write a program that accept three sides of triangle as input, and print whether the triangle is valid or not.
- Q.4) Write a Python program to check a triangle is equilateral, isosceles or scalene.
Hint: Input is the angles of triangles.
- Q.5) Write a program to check whether given character is a digit or a character in lowercase or uppercase alphabet. Hint: ASCII value of digit is between 48 to 57 and Lowercase characters in the range of 97 to 122, uppercase is between 65 and 90).
- Q.6) Write a program, which accepts annual basic salary of an employee and calculates and displays the Income tax as per the following rules.
- If Basic is less than Rs. 1,50,000/-, then Tax = 0
 - If Basic is from Rs. 1,50,000/- to Rs. 3,00,000/-, then tax is 20%
 - If Basic is greater than Rs. 3,00,000/-, then tax is 30%
- Q.7) Write a program in Python to print odd numbers from 1 to 10 except 7. (Hint: use continue statement)
- Q.8) Write a program in Python to calculate the sum of numbers in the range from m to n which are not divisible by 2,3 or 5. where $m < n$.
- Q.9) Write a program in Python to input a sentence with digits also. The program should count the number of uppercase letters, lowercase letters and digits in the sentence.
- Q.10) Write a program to find the factorial of a given positive integer n. The factorial of a number is the product of all the integers from 1 to n.
i.e., $n! = n * (n-1) * \dots * 1$.
Sample input and output
Enter the value of n : 10
Output: (10 !) = 3628800
- Q.11) A numeric palindrome is a number which remains the same even when you reverse the digits of the given number, e.g., 121, 34543, 6776, etc.
Write a Python program to check a given positive number is a numeric palindrome or not.
Sample input and output:
Input: Enter a positive integer : 12321
Output: Yes, 12321 is a numeric palindrome !!!
Input: Enter a positive integer : 96596598
Output: No, 96596598 is not a numeric palindrome !!!

- Q.12) Write a program to convert a binary number to a decimal number without using built-in function provided by Python. Note that you need to check the input is binary number or not.

Sample input and output:

Input: Enter a binary number (with 0 and 1's) : 101101

Output: The decimal number is 45

Input: Enter a binary number (with 0 and 1's) : 103402

Output: Invalid input !!!

Input: Enter a binary number (with 0 and 1's) : 10001001

Output: The decimal number is 137

- Q.13) Write a program to find out the highest common factor (HCF) of two given positive integers n and m. Then use HCF of n and m to find the least common multiple (LCM) of n and m.

Sample input and output:

Input: Enter n: 10

Enter m: 25

Output : HCF of 10 and 25 is 5

LCM of 10 and 25 is 50

Input: Enter n: 20

Enter m: 27

HCF of 20 and 27 is 1

LCM of 20 and 27 is 540

- Q.14) Write a program for the following problem.

Let N be a positive integer (read as the input). Verify that the English calendar year with year number N is a leap year or not. Note that you are not allowed to look into the calendar of the year. Just, with the information of year number you need to answer that the year is a leap or not.

Note: A year number N is said to be a leap year in the following cases: If N is a century year (e.g., 100, 200, 300, ...) and divisible by 400, then N is a leap year. If N is a non-century year and divided by 4, then N is a leap year. In all other cases, N is a non-leap year.

Sample input output:

Input: Enter a year number : 300

Output: The year 300 is not a leap year

Input: Enter a year number : 800

Output: The year 800 is a leap year

Input: Enter a year number : 2019

Output: The year 2019 is not a leap year

Input: Enter a year number : 2020

Output: The year 2020 is a leap year

- Q.15) A banking company sends a 6-digit integer number as one-time password (OTP) to its customer mobile for all mobile banking transactions. An integer is a valid OTP if (1) it is of six-digit length and (2) contains the same number of even and odd digits. Further, the OTP does not start with 0. Write a python program to read an integer and verify whether it is a valid OTP or not?

Sample input and output:

Input : Enter a number : 4564756

Output: Invalid OTP

Input: Enter a number : 676876

Output: Invalid OTP

Input: Enter a number : 123456

Output: Valid OTP